📌 BACKUP

✅ 1. backup.sh Script

Create a new file named backup.sh inside your project folder:

```
  GNU nano 7.2                                                     backup.sh
#!/bin/bash

# Create backup directory if it doesn't exists

mkdir -p backup

# Get current timestamp
timestamp=$(date +"%Y%m%d_%H%M%S")

# Find and copy .txt files with timestamp
for file in *.txt; do
 if [[ -f "$file" ]]; then
  filename=$(basename "$file" .txt)
  cp "$file" "backup/${filename}_$timestamp.txt"
  echo "Backed up: $file -> backup/${filename}_$timestamp.txt"
fi
done
```

✅ 2. Make Script Executable

Run the following command once:

chmod 777 backup.sh

✅ 3. Testing the script

1. Create some samples .txt files:

```
ashishkumar@ashishkumar:~/UPES/linux_lab$ echo "hello world" > file1.txt
ashishkumar@ashishkumar:~/UPES/linux_lab$ echo "another world" > world.txt
```

2. Run the script:

./backup.sh

```
ashishkumar@ashishkumar:~/UPES/linux_lab$ ./backup.sh
Backed up: file1.txt -> backup/file1_20250909_144443.txt
Backed up: world.txt -> backup/world_20250909_144443.txt
ashishkumar@ashishkumar:~/UPES/linux_lab$
```

3. Check the backup/folder:

ls backup/

```
ashishkumar@ashishkumar:~/UPES/linux_lab$ ls backup
file1_20250909_144219.txt  file1_20250909_144411.txt  file1_20250909_144443.txt  world_20250909_144443.txt
ashishkumar@ashishkumar:~/UPES/linux_lab$
```

🔧 LAB4– File & Backup Automation

Objective

Automate the backup of `.txt` files into a `backup/` folder with timestamps in filenames.

🔧 Script Explanation

1. mkdir -p backup
Creates a folder n".

- 🔹 3. Using at (One-time Scheduling) Run a script once at a specific time:

echo "/home/user/backup.sh" " amed backup if it does not exist.

2. timestamp=$(date +"%Y%m%d_%H%M%S")
Generates a timestamp (format: YYYYMMDD_HHMMSS).

3. for file in *.txt; do ... done
Loops through all .txt files in the current directory.

4. basename "$file" .txt
Extracts the file name without extension.

5. cp "$file" "backup/${filename}_$timestamp.txt"
Copies the file into backup/ with the timestamp appended.

🔧 Example Run

Input

Created two .txt files:

file1.txt world.txt

Command

./backup.sh

Output

Files copied into backup/ with timestamps:

```
ashishkumar@ashishkumar:~/UPES/linux_lab$ ./backup.sh
Backed up: file1.txt -> backup/file1_20250909_200925.txt
Backed up: world.txt -> backup/world_20250909_200925.txt
ashishkumar@ashishkumar:~/UPES/linux_lab$
```

🛠️ Q1-What is the difference between cp,mv,and rsync?

```
ans=cp-Copies files or directories
    rm-Moves or renames files or directories
    rsync-Synchronizes files/directories efficiently
```

✅ Q2-How can you schedule scripts to run automatically?

```
 ans=You can schedule scripts to run automatically using task schedulers
built into your operating system.
```